

Power Web Discovery TO BE 0.7.6.6.2

Review design for a planned Radar search pipeline change

Status: Implemented.

Pipeline id: power-web-discovery.

AS IS: ../RADAR_POWER_WEB_DISCOVERY_AS_IS.md

Baseline: ../validation/0.7.6.6.2/BASELINE_DIAGNOSTIC.md

Acceptance manifest: RADAR_POWER_WEB_DISCOVERY_TO_BE_0.7.6.6.2.acceptance.json

Implemented stage: people-search-stage-9b8e0bac39a759b264af.

Validation: ../validation/0.7.6.6.2/VALIDATION_REPORT.md (PASS).

Intent

Turn one immutable power_web_handoff.v1 into an auditable people-source search stage without creating identities, employment claims or graph edges:

text

handoff -> planning input -> title hypotheses -> deterministic acceptance

-> lane strategy -> bounded retrieval -> source leads

-> role coverage checkpoints -> people_search_stage.v1

The stage writes a product-safe JSON artifact and validation report. Persisted power-web-run-*, APIs, jobs, history and product UI remain owned by later slices.

Ownership and boundaries

The Power Web application package owns planning input, hypothesis acceptance, source obligations, budget decisions, receipt semantics, lead normalization, coverage checkpoints and artifact projection. Integrations own OpenRouter HTTP transport and response normalization. CLI composition loads the existing handoff and writes an explicit artifact file.

Application code imports no candidate-discovery or signal-monitoring internals, FastAPI, SQLAlchemy, Celery, HTTP client or provider SDK. It consumes the existing handoff and source evidence through provider-neutral ports. Existing AccountRoleTitleHypothesis remains a compatibility contract; runtime uses proposal and accepted-hypothesis records with exact demand/product/version lineage.

Planning and hypothesis acceptance

The input contains the immutable handoff, account aliases, verified official domains with product-safe source evidence refs, language/geography context,

capability cards and the execution-profile snapshot. A bare domain without provenance is invalid. The input never contains blind controls.

One batch provider call proposes at most five title/function variants per role; one schema retry is allowed. The deterministic acceptance service retains at most three normalized variants per role and records every rejection. A proposal cannot add or remove RoleDemand, change product lineage, requiredness, priority or scope. Person names, contacts, URLs, benchmark controls, empty values, duplicates and unrelated variants are rejected. A deterministic fallback from display name and responsibility guarantees one accepted hypothesis per role. Original account language is preserved; unprovided transliteration is not invented.

Source-lane strategy

The quality profile selects three independent mandatory lanes per role when their capabilities are available:

- `official_company`: requires a verified account-owned domain;
- `hh_public_web`: normal web search restricted to hh.ru;
- `generic_web`: unrestricted public-web search.

Professional-network, publication/event, procurement/patent and industry lanes are supported as conditional decisions but disabled in the acceptance profile.

Every selected decision terminates as scheduled, executed, `not_executable`, unsupported, `policy_limited` or `budget_limited`.

Mandatory decisions may not disappear between strategy and execution.

HH access is public-search-only. The stage performs no HH API/OAuth call, authentication, direct mass crawling, CAPTCHA/robots bypass or contact extraction. An inaccessible page may remain a limited lead using its public search citation; a snippet never confirms identity or employment.

Retrieval, receipts and leads

Every executed task produces a `PeopleSearchExecutionReceipt` with task, demand, product, hypothesis and lane lineage; sanitized query and restrictions; configured provider/engine; timestamps; result count; normalized source refs; attempt outcome and retry count. Raw response, prompt, headers, credentials, HTML and hidden reasoning are never retained.

`PeopleSourceLead` stores a canonical URL, title, product-safe snippet, optional public name/title/employer/geography hints, lane, source capability and reason. It is not a `PersonProfile`, identity decision, employment claim, influence hypothesis or Power Web node.

Provider timeout/error receives one same-engine retry. It never becomes `searched_no_results`. Coverage is one of `complete_with_leads`,

complete_no_leads, incomplete_capability, incomplete_provider, incomplete_budget or incomplete_policy. complete_no_leads requires a successful receipt from every applicable mandatory lane.

Budgets

Profile people_search_quality snapshots independent settings and counters:

- 8 role demands;
- 2 hypothesis-provider calls;
- 5 proposed and 3 accepted variants per role;
- 24 initial tasks, 40 total tasks and 48 provider calls;
- 80 source verifications;
- one query revision per role/lane and one provider retry per task;
- reserves: 8 HH, 8 official, 8 generic and 16 revision/recovery tasks.

One configured OpenRouter primary engine is used. Retry does not switch engine. Candidate-discovery and Signal Monitoring budgets are neither read nor changed.

Artifact and evaluation

people_search_stage.v1 contains stage execution ID, handoff/account/product lineage, planning input summary, proposal and acceptance ledgers, source-lane ledger, tasks, attempts, receipts, leads, coverage checkpoints, model/provider metadata, budgets, diagnostics and terminal state. It is file-based and not a persisted Power Web run.

After execution the evaluator loads the frozen benchmark. It reports each blind profile control as retrieved or with a path-level miss reason while proving controls_in_planning_count=0. This slice validates retrieval coverage only; identity quality remains outside its claims.

Test and live acceptance

Recorded tests exercise all fourteen demands from the two-product handoff, malformed proposals, duplicates, policy mutation, unknown lanes, missing official domain, retries, budget exhaustion, searched-negative semantics, privacy filtering and blind leakage. The live gate uses the eight-role SmartDiagnostics handoff from the baseline diagnostic and executes exactly 24 initial mandatory-lane decisions through the remote contour.

The slice passes only with receipts for every executed task, no orphan decision or silent drop, at least one retained lead from each mandatory lane, relevant lead coverage for at least four roles, zero unrecovered mandatory-lane provider errors, provider calls within 48 and planner calls within 2. Candidate, Signal and persisted Power Web runs remain zero.

Out of scope

- No person profile extraction or cross-source identity resolution.
- No employment, relationship or influence decision.
- No database migration, API, job, run history or UI.
- No HH authorized API and no comparison of OpenRouter search engines.
- No public quality claim from this single acceptance run.

Requirement traceability

The adjacent manifest defines PW-PS-ASIS-01, PW-PS-IN-01, PW-PS-HYP-01, PW-PS-HYP-02, PW-PS-LANE-01, PW-PS-LANE-02, PW-PS-HH-01, PW-PS-AUD-01, PW-PS-NEG-01, PW-PS-BUD-01, PW-PS-SEC-01, PW-PS-BENCH-01, PW-PS-ARCH-01, PW-PS-LIVE-01 and PW-PS-PROC-01. Every mandatory requirement must pass recorded tests and the declared remote live evidence before this design becomes AS IS.